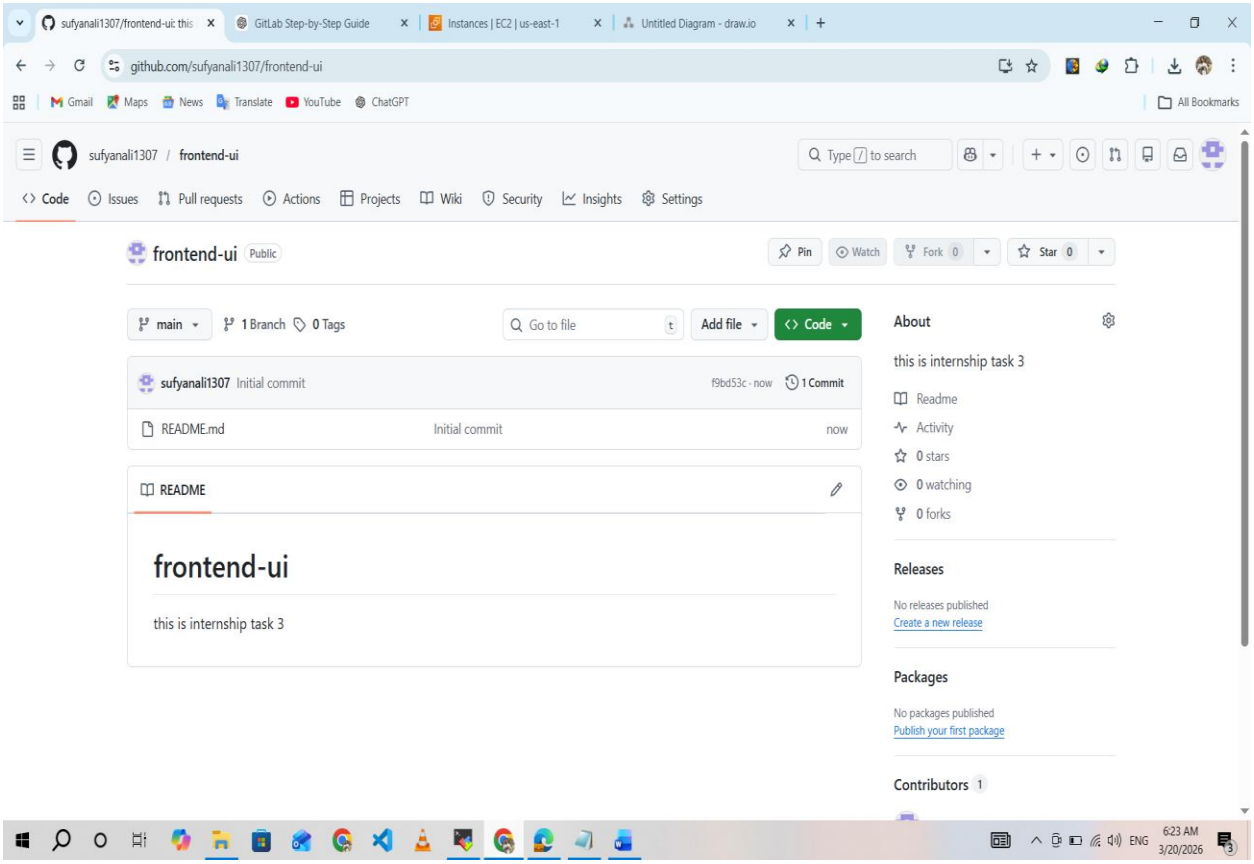
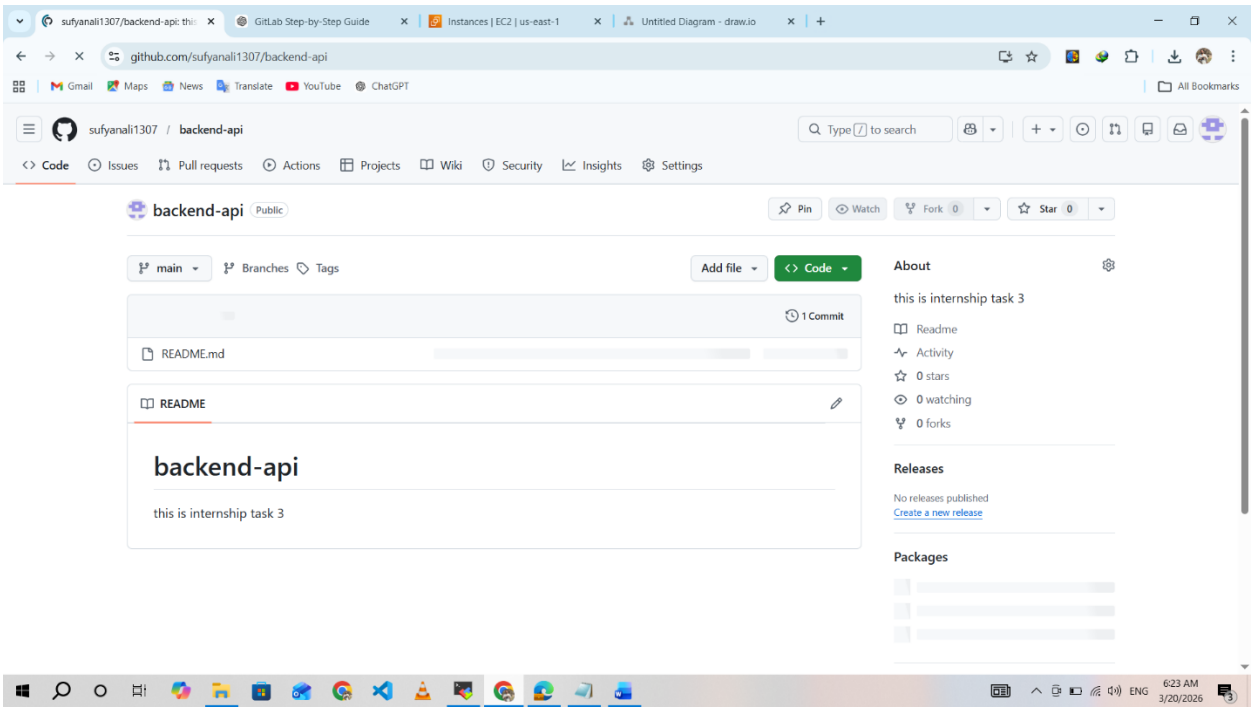
Name: Hafiz Sufyan Ali

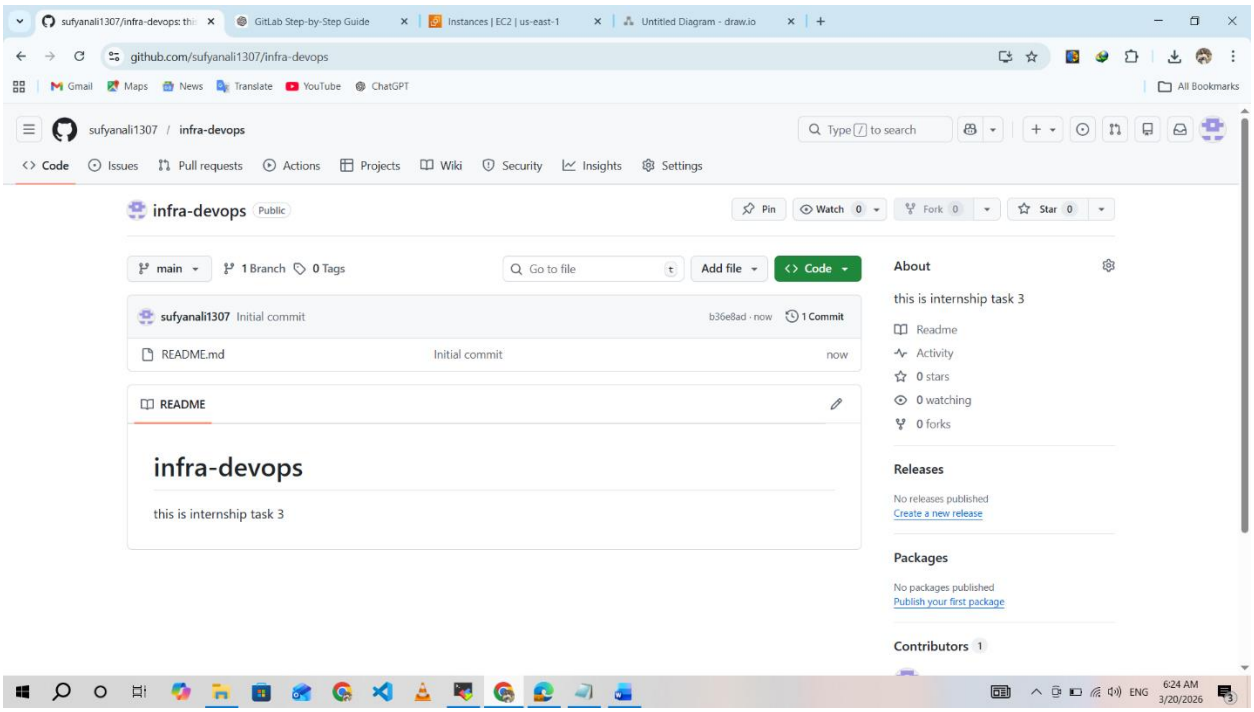
Scenario Task 2: Design Repositories for Two Different Organizations

Company A — Small Startup Create Repo Design Frontend Backend and Infra Devops

Pull and Merge Commit Pictures




sufyanali1307/backend-api: thi
GitLab Step-by-Step Guide
Instances | EC2 | us-east-1
Untitled Diagram - draw.io
github.com/sufyanali1307/backend-api
Gmail Maps News Translate YouTube ChatGPT
All Bookmarks
sufyanali1307 / backend-api
Type / to search
Code Issues Pull requests Actions Projects Wiki Security Insights Settings
backend-api Public
Pin Watch Fork 0 Star 0
main Branches Tags
Add file Code
1 Commit
README.md
README
backend-api
this is internship task 3
About
this is internship task 3
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
ENG
6:23 AM
3/20/2026

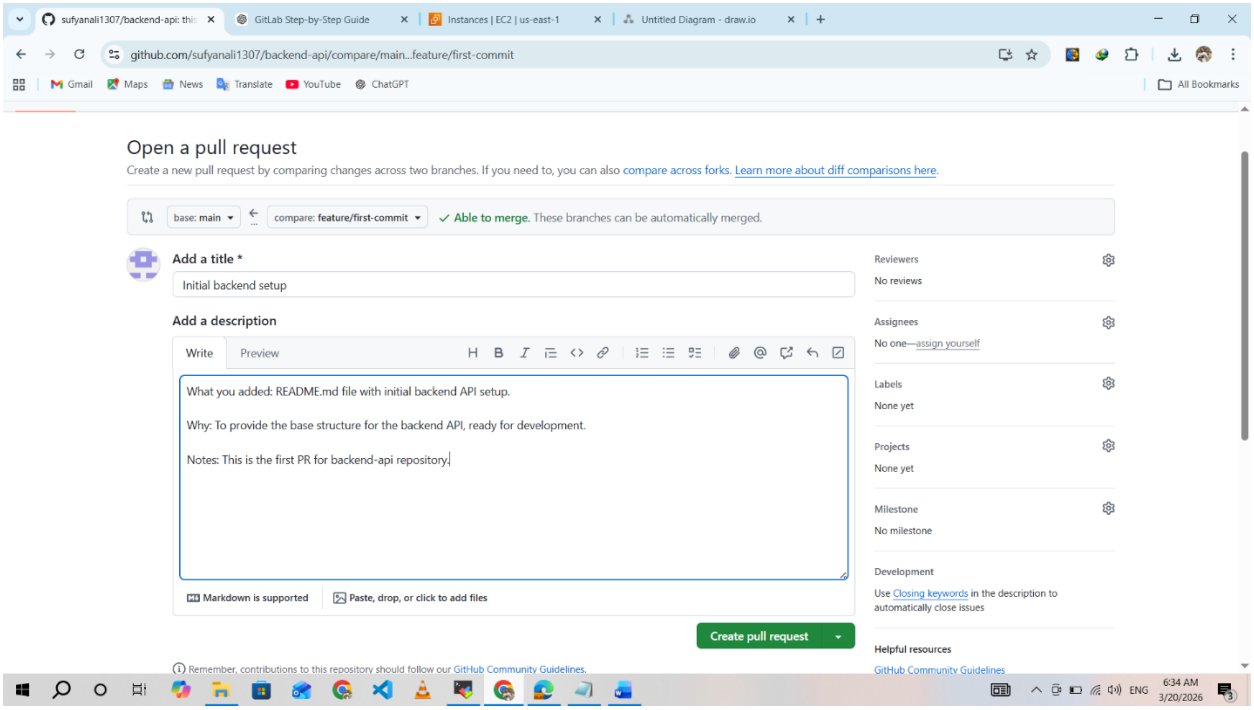

sufyanali1307/infra-devops: thi
GitLab Step-by-Step Guide
Instances | EC2 | us-east-1
Untitled Diagram - draw.io
github.com/sufyanali1307/infra-devops
Gmail Maps News Translate YouTube ChatGPT
All Bookmarks
sufyanali1307 / infra-devops
Type / to search
Code Issues Pull requests Actions Projects Wiki Security Insights Settings
infra-devops Public
Pin Watch 0 Fork 0 Star 0
main 1 Branch 0 Tags
Go to file
Add file Code
sufyanali1307 Initial commit
b36e8ad · now
1 Commit
README.md
Initial commit
now
README
infra-devops
this is internship task 3
About
this is internship task 3
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
ENG
6:24 AM
3/20/2026

Terminal Commands

```
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature$ git clone https://github.com/sufyanali1307/backend-api.git
Cloning into 'backend-api'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature$ la
.git  README.md  about-me.txt  backend-api
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature$ cd backend-api/
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ ls
README.md
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ git checkout -b feature/first-commit
Switched to a new branch 'feature/first-commit'
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ echo "Initial backend API code" >> README.md
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ goit add .
Command 'goit' not found, did you mean:
  command 'gost' from snap gost (2.12.0)
  command 'doit' from deb python3-doit (0.31.1-3.4)
  command 'got' from deb got (0.95-1)
  command 'gost' from deb gost (0.1.2-2ubuntu0.24.04.3)
  command 'git' from deb git (1:2.43.0-1ubuntu7.3)
  command 'gout' from deb scotch (7.0.4-1)
See 'snap info <snapname>' for additional versions.
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ git add .
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ git commit -m "Initial backend API setup"
[feature/first-commit 4ddfaad] Initial backend API setup
 1 file changed, 1 insertion(+)
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$
```

```
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ git checkout -b feature/first-commit
Switched to a new branch 'feature/first-commit'
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ echo "Initial backend API code" >> README.md
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ goit add .
Command 'goit' not found, did you mean:
  command 'gost' from snap gost (2.12.0)
  command 'doit' from deb python3-doit (0.31.1-3.4)
  command 'got' from deb got (0.95-1)
  command 'gost' from deb gost (0.1.2-2ubuntu0.24.04.3)
  command 'git' from deb git (1:2.43.0-1ubuntu7.3)
  command 'gout' from deb scotch (7.0.4-1)
See 'snap info <snapname>' for additional versions.
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ git add .
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ git commit -m "Initial backend API setup"
[feature/first-commit 4ddfaad] Initial backend API setup
 1 file changed, 1 insertion(+)
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$ git push origin feature/first-commit
Username for 'https://github.com': sufyanali1307
Password for 'https://sufyanali1307@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 316 bytes | 316.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature/first-commit' on GitHub by visiting:
remote:      https://github.com/sufyanali1307/backend-api/pull/new/feature/first-commit
remote:
To https://github.com/sufyanali1307/backend-api.git
 * [new branch]      feature/first-commit -> feature/first-commit
ubuntu@ip-172-31-84-181:~/devops-week1-git-lab/team-devops-profile-feature/backend-api$
```

Pull request



Initial backend setup by sufyan | GitLab Step-by-Step Guide | Instances | EC2 | us-east-1 | Untitled Diagram - draw.io

github.com/sufyanali1307/backend-api/pull/1

Gmail Maps News Translate YouTube ChatGPT All Bookmarks

Code Issues Pull requests 1 Actions Projects Wiki Security Insights Settings

Initial backend setup #1

Ready to merge Code

Open sufyanali1307 wants to merge 1 commit into `main` from `feature/first-commit`

Conversation 0 Commits 1 Checks 0 Files changed 1 +1

sufyanali1307 commented now Owner

What you added: README.md file with initial backend API setup.

Why: To provide the base structure for the backend API, ready for development.

Notes: This is the first PR for backend-api repository.

`Initial backend API setup` 4ddfaad

No conflicts with base branch
Merging can be performed automatically.

Merge pull request You can also merge this with the command line. View command line instructions.

Still in progress? Convert to draft

Reviewers
No reviews
Still in progress? Convert to draft

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.

ENG 6:34 AM 3/20/2026

Initial backend setup by sufyan | GitLab Step-by-Step Guide | Instances | EC2 | us-east-1 | Untitled Diagram - draw.io

github.com/sufyanali1307/backend-api/pull/1

Gmail Maps News Translate YouTube ChatGPT All Bookmarks

Merged **Initial backend setup** #1
sufyanali1307 merged 1 commit into `main` from `feature/first-commit`

Notes: This is the first PR for backend-api repository.

`Initial backend API setup` 4ddfaad

sufyanali1307 merged commit `c976400` into `main` now Revert

Pull request successfully merged and closed
You're all set — the `feature/first-commit` branch can be safely deleted. Delete branch

Add a comment

Write Preview

Add your comment here...

Markdown is supported Paste, drop, or click to add files

Comment

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

Notifications Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.

1 participant

ENG 6:35 AM 3/20/2026

```
ubuntu@ip-172-31-84-181:~$ git clone https://github.com/sufyanali1307/frontend-ui.git
Cloning into 'frontend-ui'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
ubuntu@ip-172-31-84-181:~$ ls
archive.tar  devops-week1-git-lab  file.log  file1.txt  file2.txt  file2.zip  folder  frontend-ui
ubuntu@ip-172-31-84-181:~$ cd frontend-ui/
ubuntu@ip-172-31-84-181:~/frontend-ui$ git checkout -b feature/first-frontend
Switched to a new branch 'feature/first-frontend'
ubuntu@ip-172-31-84-181:~/frontend-ui$ echo "Initial frontend UI setup" >> README.md
ubuntu@ip-172-31-84-181:~/frontend-ui$ git add .
git commit -m "Initial frontend UI setup"
[feature/first-frontend 380762d] Initial frontend UI setup
 1 file changed, 1 insertion(+)
ubuntu@ip-172-31-84-181:~/frontend-ui$ git push origin feature/first-frontend
Username for 'https://github.com': sufyanali1307
Password for 'https://sufyanali1307@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 315 bytes | 315.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature/first-frontend' on GitHub by visiting:
remote:      https://github.com/sufyanali1307/frontend-ui/pull/new/feature/first-frontend
remote:
To https://github.com/sufyanali1307/frontend-ui.git
 * [new branch]      feature/first-frontend -> feature/first-frontend
ubuntu@ip-172-31-84-181:~/frontend-ui$
```

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also compare across forks. Learn more about diff comparisons here.

base: main ← compare: feature/first-frontend ✓ Able to merge. These branches can be automatically merged.

Add a title *

Initial frontend UI setup

Add a description

Write | Preview

What you added: README.md file with initial frontend structure.

Why: To provide the base structure for frontend development.

Notes: This is the first PR for frontend-ui repository.

Markdown is supported | Paste, drop, or click to add files

Create pull request

Reviewers: No reviews

Assignees: No one—assign yourself

Labels: None yet

Projects: None yet

Milestone: No milestone

Development: Use Closing keywords in the description to automatically close issues

Initial frontend UI setup by sufy | GitLab Step-by-Step Guide | Instance details | EC2 | us-east- | Untitled Diagram - draw.io

github.com/sufyanali1307/frontend-ui/pull/1

Gmail Maps News Translate YouTube ChatGPT All Bookmarks

Initial frontend UI setup #1

Ready to merge | Code

Open sufyanali1307 wants to merge 1 commit into `main` from `feature/first-frontend`

Conversation 0 | Commits 1 | Checks 0 | Files changed 1

sufyanali1307 commented now — Owner

What you added: README.md file with initial frontend structure.

Why: To provide the base structure for frontend development.

Notes: This is the first PR for frontend-ui repository.

Initial frontend UI setup 380762d

No conflicts with base branch
Merging can be performed automatically.

Merge pull request — You can also merge this with the command line. View command line instructions.

Still in progress? Convert to draft

Add a comment

Reviewers: No reviews
Still in progress? Convert to draft

Assignees: No one—assign yourself

Labels: None yet

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues.
None yet

ENG 7:05 PM 3/20/2026

Initial frontend UI setup by sufy | GitLab Step-by-Step Guide | Instance details | EC2 | us-east- | Untitled Diagram - draw.io

github.com/sufyanali1307/frontend-ui/pull/1

Gmail Maps News Translate YouTube ChatGPT All Bookmarks

Merged **Initial frontend UI setup** #1
sufyanali1307 merged 1 commit into `main` from `feature/first-frontend`

Initial frontend UI setup 380762d

sufyanali1307 merged commit `adda9fa` into `main` now — Revert

Pull request successfully merged and closed
You're all set — the `feature/first-frontend` branch can be safely deleted. — Delete branch

Add a comment

Write | Preview

Add your comment here...

Markdown is supported | Paste, drop, or click to add files

Comment

Remember, contributions to this repository should follow our GitHub Community Guidelines.

ProTip! Add comments to specific lines under Files changed.

Labels: None yet

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues.
None yet

Notifications — Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.

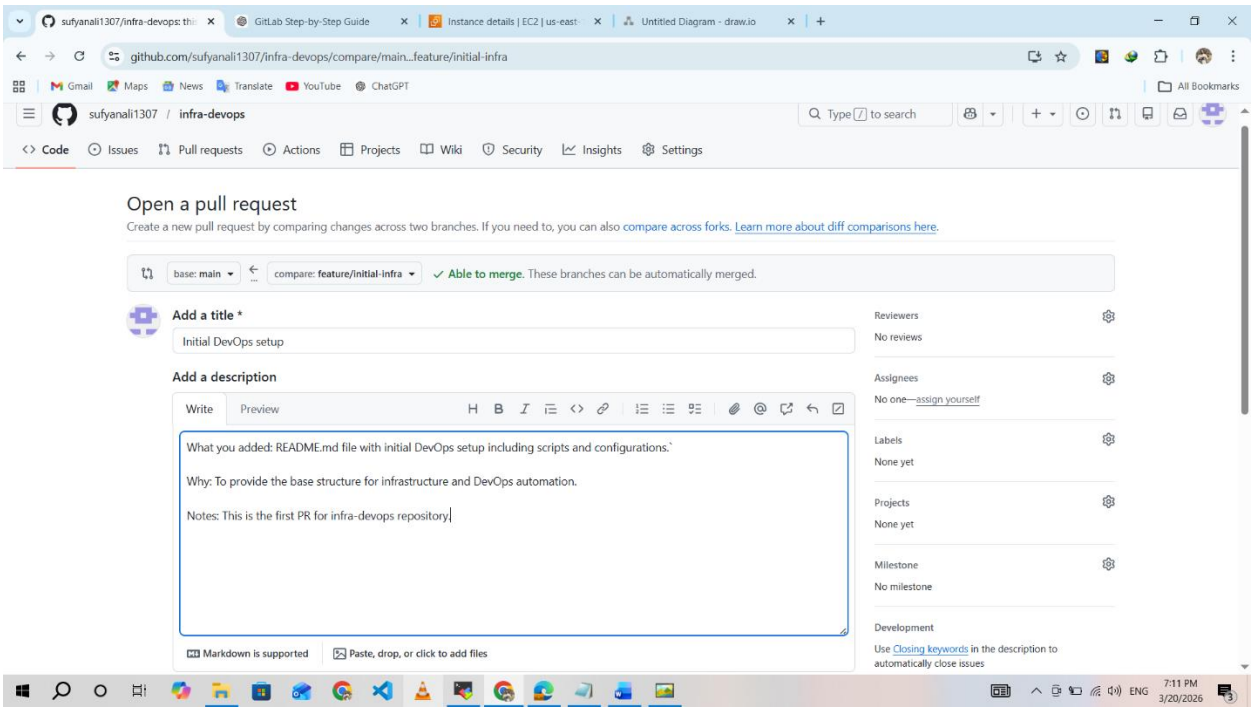
1 participant

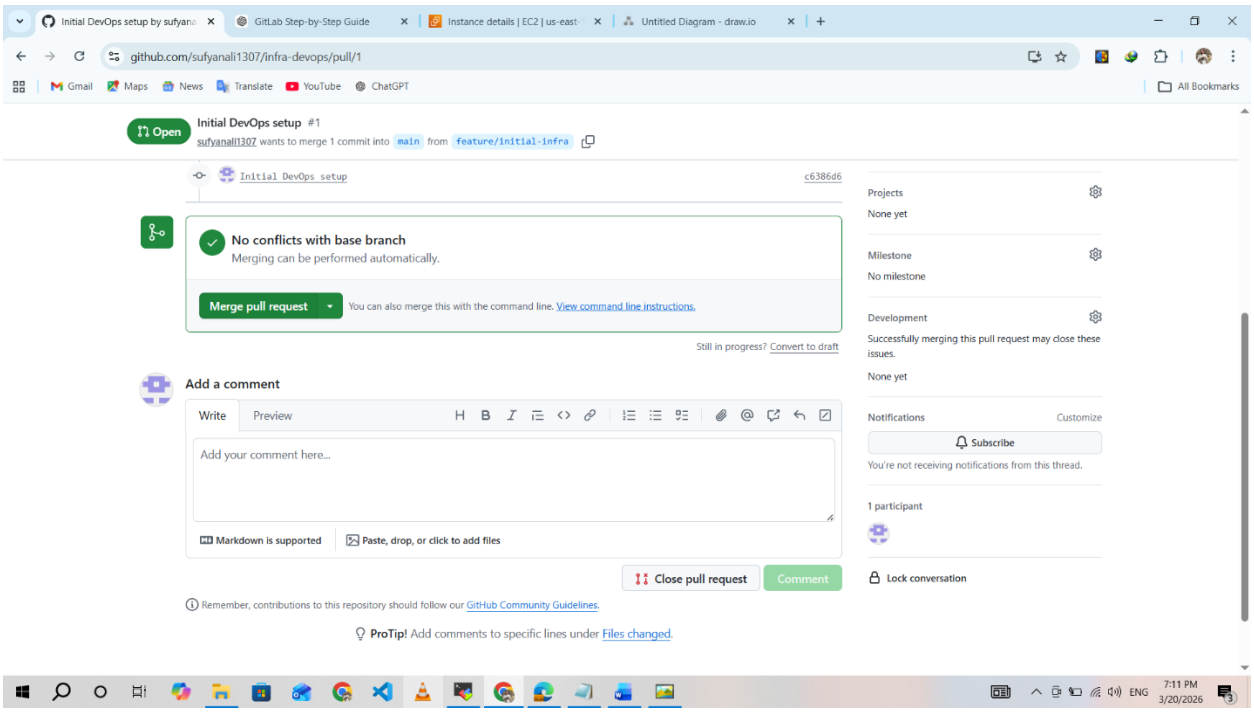
Lock conversation

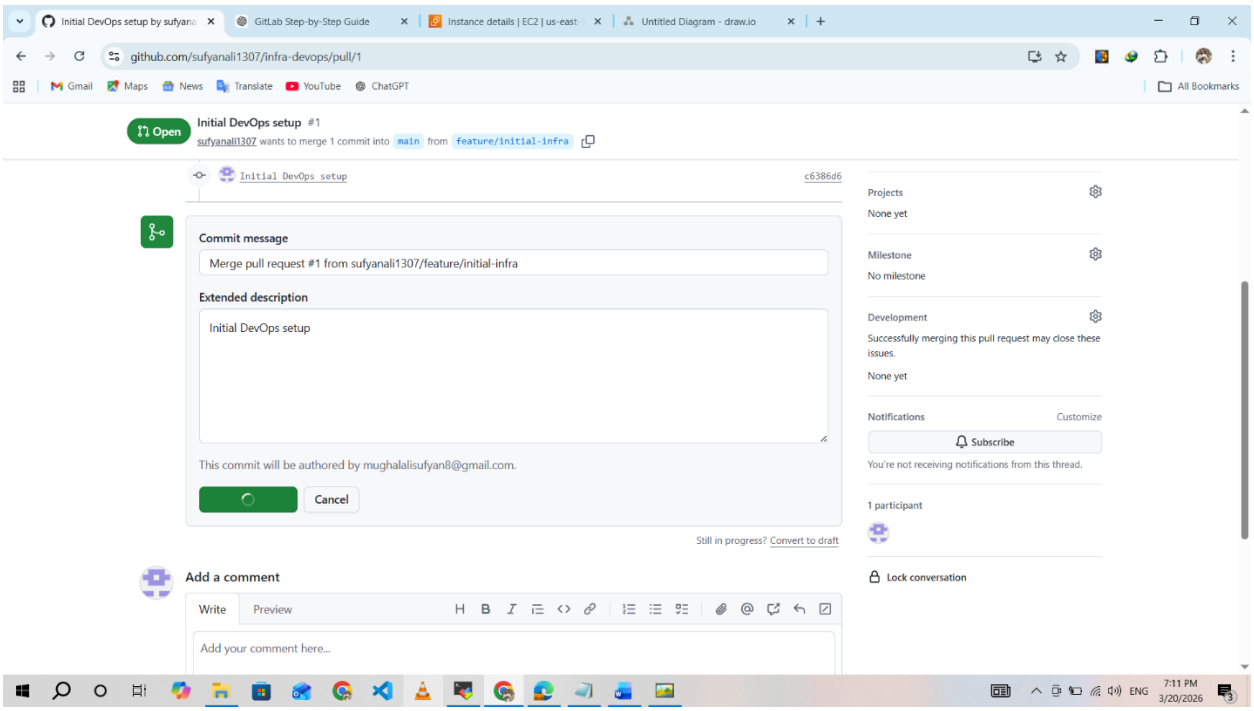
ENG 7:05 PM 3/20/2026

```
ubuntu@ip-172-31-84-181:~/frontend-ui$ git clone https://github.com/sufyanali1307/infra-devops.git
Cloning into 'infra-devops'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
ubuntu@ip-172-31-84-181:~/frontend-ui$ ls
README.md  infra-devops
ubuntu@ip-172-31-84-181:~/frontend-ui$ cd infra-devops/
ubuntu@ip-172-31-84-181:~/frontend-ui/infra-devops$ cd ..
ubuntu@ip-172-31-84-181:~/frontend-ui$ cd ..
ubuntu@ip-172-31-84-181:~$ git clone https://github.com/sufyanali1307/infra-devops.git
Cloning into 'infra-devops'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
ubuntu@ip-172-31-84-181:~$ ls
archive.tar  devops-week1-git-lab  file.log  file1.txt  file2.txt  file2.zip  folder  frontend-ui  infra-devops
ubuntu@ip-172-31-84-181:~$ cd infra-devops/
ubuntu@ip-172-31-84-181:~/infra-devops$ ls
README.md
ubuntu@ip-172-31-84-181:~/infra-devops$ git checkout -b feature/initial-infra
Switched to a new branch 'feature/initial-infra'
ubuntu@ip-172-31-84-181:~/infra-devops$ echo "Initial DevOps scripts and setup" >> README.md
ubuntu@ip-172-31-84-181:~/infra-devops$ git add .
git commit -m "Initial DevOps setup"
[feature/initial-infra c6386d6] Initial DevOps setup
 1 file changed, 1 insertion(+)
ubuntu@ip-172-31-84-181:~/infra-devops$ git push origin feature/initial-infra
Username for 'https://github.com': sufyanali1307
Password for 'https://sufyanali1307@github.com':
```

```
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
ubuntu@ip-172-31-84-181:~$ ls
archive.tar  devops-week1-git-lab  file.log  file1.txt  file2.txt  file2.zip  folder  frontend-ui  infra-devops
ubuntu@ip-172-31-84-181:~$ cd infra-devops/
ubuntu@ip-172-31-84-181:~/infra-devops$ ls
README.md
ubuntu@ip-172-31-84-181:~/infra-devops$ git checkout -b feature/initial-infra
Switched to a new branch 'feature/initial-infra'
ubuntu@ip-172-31-84-181:~/infra-devops$ echo "Initial DevOps scripts and setup" >> README.md
ubuntu@ip-172-31-84-181:~/infra-devops$ git add .
git commit -m "Initial DevOps setup"
[feature/initial-infra c6386d6] Initial DevOps setup
 1 file changed, 1 insertion(+)
ubuntu@ip-172-31-84-181:~/infra-devops$ git push origin feature/initial-infra
Username for 'https://github.com': sufyanali1307
Password for 'https://sufyanali1307@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 326 bytes | 326.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature/initial-infra' on GitHub by visiting:
remote:      https://github.com/sufyanali1307/infra-devops/pull/new/feature/initial-infra
remote:
To https://github.com/sufyanali1307/infra-devops.git
 * [new branch]      feature/initial-infra -> feature/initial-infra
ubuntu@ip-172-31-84-181:~/infra-devops$
```


sufyanali1307/infra-devops: thi
GitLab Step-by-Step Guide
Instance details | EC2 | us-east
Untitled Diagram - draw.io
github.com/sufyanali1307/infra-devops/compare/main...feature/initial-infra
Gmail Maps News Translate YouTube ChatGPT
All Bookmarks
sufyanali1307 / infra-devops
Type / to search
Code Issues Pull requests Actions Projects Wiki Security Insights Settings
Open a pull request
Create a new pull request by comparing changes across two branches. If you need to, you can also compare across forks. Learn more about diff comparisons here.
base: main
compare: feature/initial-infra
Able to merge. These branches can be automatically merged.
Add a title *
Initial DevOps setup
Add a description
Write Preview
What you added: README.md file with initial DevOps setup including scripts and configurations.`
Why: To provide the base structure for infrastructure and DevOps automation.
Notes: This is the first PR for infra-devops repository.
Markdown is supported
Paste, drop, or click to add files
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Use Closing keywords in the description to automatically close issues
ENG
7:11 PM
3/20/2026


Initial DevOps setup by sufyana
GitLab Step-by-Step Guide
Instance details | EC2 | us-east
Untitled Diagram - draw.io
github.com/sufyanali1307/infra-devops/pull/1
Gmail Maps News Translate YouTube ChatGPT
All Bookmarks
Open
Initial DevOps setup #1
sufyanali1307 wants to merge 1 commit into main from feature/initial-infra
Initial DevOps setup
c6386d6
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.
Still in progress? Convert to draft
Add a comment
Write Preview
Add your comment here...
Markdown is supported
Paste, drop, or click to add files
Close pull request
Comment
Remember, contributions to this repository should follow our GitHub Community Guidelines.
ProTip! Add comments to specific lines under Files changed.
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Subscribe
You're not receiving notifications from this thread.
1 participant
Lock conversation
ENG
7:11 PM
3/20/2026



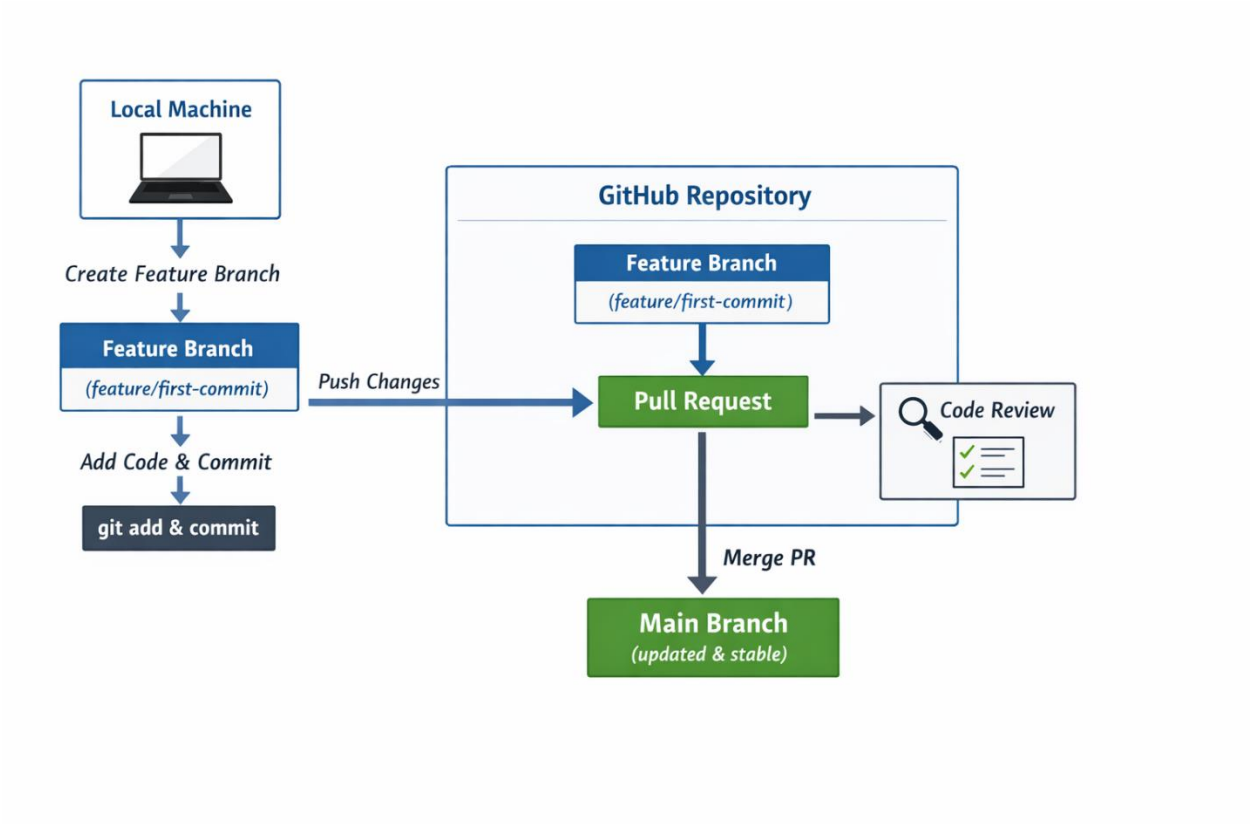
Links Pull links

https://github.com/sufyanali1307/backend-api/pull/1

https://github.com/sufyanali1307/frontend-ui/pulls

https://github.com/sufyanali1307/infra-devops/pull/1

Diagram



Summary

The developer creates a feature branch on the local machine, makes changes, and pushes it to GitHub. A Pull Request is created to merge the feature branch into the main branch. After review, the PR is merged, updating the main branch while keeping it stable and clean.

Company B — Medium Organization

Repository Design

For the small startup, I created three separate repositories to manage different parts of the application:

- backend-Api
- frontend-UI
- infra-devops

Repository Purpose

- **backend-api:** Contains backend application code and logic
- **frontend-ui:** Contains frontend user interface code
- **infra-devops:** Contains DevOps configurations like Docker and deployment setup

Branching Strategy

A simple branching strategy is used:

- main → stable production code
- feature/* → for new features

Workflow Explanation

Developers create feature branches, make changes, push code to GitHub, and create Pull Requests. After review, the changes are merged into the main branch.

Company B — Medium Organization

Number of Repositories

For the medium organization, multiple repositories are used to manage microservices and infrastructure separately:

- user-service

- order-service
- payment-service
- infra-terraform
- shared-libs

Microservice Repository Structure

Each microservice has its own repository to ensure code isolation and independent development.

Example:

```
user-service/
 ├── src/
 ├── tests/
 ├── Dockerfile
 ├── README.md
 └── .github/workflows/ci.yml
```

Infrastructure Repository

A separate repository named infra-terraform is used to manage infrastructure using Terraform.

Branching Strategy

A structured branching strategy is followed:

- main → production
- develop → integration
- feature/* → new features
- hotfix/* → urgent fixes

Branch Protection Rules

The main branch is protected with the following rules:

- No direct commits allowed
- Pull Request required before merge

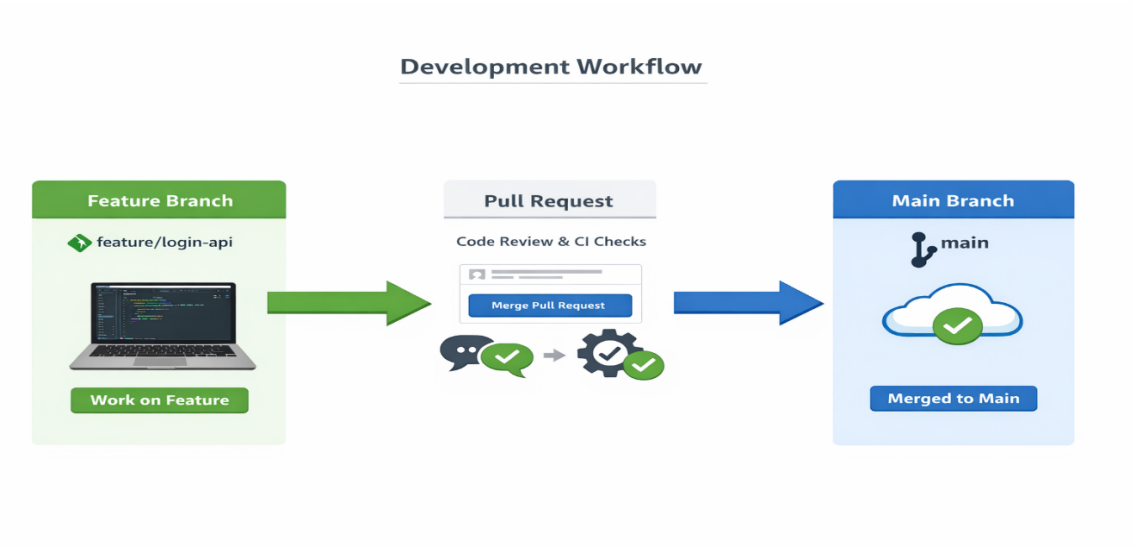
- At least one reviewer approval required
- CI/CD checks must pass before merging

GitHub Policies

The following GitHub policies are enabled:

1. Require Pull Request before merge
2. Require at least 1 reviewer
3. Enable secret scanning
4. Require CI checks before merge

Diagram



Workflow Explanation

Developers work on feature branches, push code to GitHub, and create Pull Requests. After code review and successful CI checks, the changes are merged into the main branch. This ensures code quality, collaboration, and stability.